PrivacyGuard

RL for adaptive data-sharing decisions

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Abstract. PrivacyGuard learns when to share full data, provide a coarse response, ask for consent, or deny access. A finite exposure budget and consent fatigue give decisions delayed consequences. We implement tabular Q-learning with a hard permission/budget guard and evaluate eight independent runs against SARSA, planning/observation ablations, reference strategies, and an analytical ceiling. On held-out synthetic sessions, Q-learning improves return by 35.26% over greedy sharing and attains 95.98% of the model optimum. A saved-policy demo exposes Q values, training counts, and session audits.

1 Problem and sequential motivation

An assistant handling location, spending, or activity requests must balance task value with disclosure. A low-value disclosure now may consume budget needed later; repeated consent prompts can reduce later acceptance. This is sequential control without labelled correct actions. Context-sensitive handling is motivated by contextual integrity [3]; our numerical costs are explicit synthetic assumptions.

Project contribution. We integrate exposure allocation, stochastic consent, action-induced fatigue, and permission enforcement in a small explainable MDP. Multiseed ablations, an exact diagnostic, and an offline demo provide an auditable evaluation. We do not claim a new RL algorithm or established research novelty.

2 RL formulation and cost model

State space. The six-component observation is $s = (h, b, f, u, d, p)$: requests remaining $h \in \{0, \dots, 12\}$, exposure budget $b \in \{0, \dots, 10\}$, consent fatigue $f \in \{0, 1, 2\}$, utility and sensitivity indices $u, d \in \{0, 1, 2\}$, and standing full-permission bit $p \in \{0, 1\}$. There are $13 \cdot 11 \cdot 3 \cdot 3 \cdot 3 \cdot 2 = 7{,}722$ combinations including terminal states, 7,128 nonterminal combinations, and 30,888 Q entries. Sessions start at $(h, b, f) = (12, 10, 0)$ and end at $h = 0$.

Request dynamics. Utility is $U = (1, 3, 7)_u$ and full exposure is $E = (1, 2, 4)_d$. New contexts are independent draws with marginals $P(u) = P(d) = (.45, .35, .20)$ and $P(p) = (.40, .60)$. With realised exposure e, $b' = b - e$ and $h' = h - 1$. Asking gives $f' = \min(2, f + 1)$; any other action gives $f' = \max(0, f - 1)$. This observation contains the history needed by the assumed dynamics, satisfying the Markov property.

Actions and hard guard. $\mathcal{A} = \{\text{deny}, \text{coarse}, \text{ask}, \text{full}\}$. Full requires $p = 1$ and $b \geq E$; coarse requires $b \geq \lceil .35E \rceil$; ask requires $b \geq E$ so acceptance can be honoured. Deny is always feasible. Coarse sharing assumes an existing limited permission. The same guard applies to all learners, references, exploration, and evaluation. It enforces feasibility; RL ranks the remaining choices.

Stochastic consent. Asking obtains one-request full permission with probability

$$P(z = 1 \mid d, f) = \text{clip}(.92 - .12d - .25f, .10, .95).$$

Rejection shares nothing. Accepted asking and direct full sharing spend the *same* exposure: consent does not erase information cost. Fatigue increases after either consent outcome.

Reward. For realised utility v and exposure e:

$$(v, e) = \begin{cases} (0, 0) & \text{deny or rejected ask,} \\ (.60U, \lceil .35E \rceil) & \text{coarse,} \\ (U, E) & \text{full or accepted ask,} \end{cases}$$

$$r = v - .45e - (.25 + .30f)\mathbf{1}_{a=\text{ask}} - .15U\mathbf{1}_{v=0}.$$

Utility rewards service; exposure penalises disclosure; prompts price interruption; missed utility prevents deny-all from being cost-free. At $U = 7, E = 4, f = 0$, rewards are 5.20 (full), 3.30 (coarse), 4.95 (accepted ask), -1.30 (rejected ask), and -1.05 (deny). No reward overrides permission.

3 Learning and experimental design

Algorithm. Our sample-based implementation of Q-learning [1, 2] updates

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \max_{a' \in \mathcal{A}(s')} Q(s', a') - Q(s, a) \right].$$

We use $\gamma = 1$ for the finite horizon and zero terminal future value. All Q values start at zero. Each method receives 120,000 episodes per seed. Epsilon decays from 1 to .05 over the first 80% of training. For post-increment visit count N, $\alpha = \max(.04, .70/(1 + N/80)^{.60})$; its floor means no asymptotic convergence is claimed. To cover rare states, 20% of episodes use uniformly sampled nonterminal (h, b, f) starts. All test sessions start normally.

Controls. SARSA uses its next epsilon-greedy action in the target. Myopic Q uses $\gamma = 0$. Budget-blind/fatigue-blind Q omit the named observation but retain the guard; action feasibility therefore leaks some budget information. Simple references are immediate-reward greedy, always coarse, full-or-consent, always consent, deny-all, and a fixed random feasible lookup. The primary submitted solution is RL. Separate known-model backward induction provides an optimal-return ceiling and simulator check; it never supplies learner targets, initial values, or action labels.

Protocol. Training uses seeds 0–7; validation uses seed 50,000 (384 sessions); held-out testing uses 90,000–90,007 (5,000 sessions each). Within each test seed, all policies share request contexts and exogenous consent uniforms. Final checkpoints are evaluated without choosing the strongest checkpoint/seed. The demo agent is preselected seed 0. Means and 95% Student-t intervals use eight independent seed means, not 40,000 sessions as independent training runs.

Update audit. Each step masks actions, samples an epsilon-greedy choice, draws the consent outcome and next request, then updates only the visited Q entry. The TD target uses realised transitions, never the exact model expectation. Across 40 runs, 4.8 million training episodes produce 52.3 million sampled transitions. Checkpoints retain both Q values and visit counts; a separate audit recomputes all 96 nominal policy/seed evaluations and verifies checkpoint hashes and browser/Python action agreement.

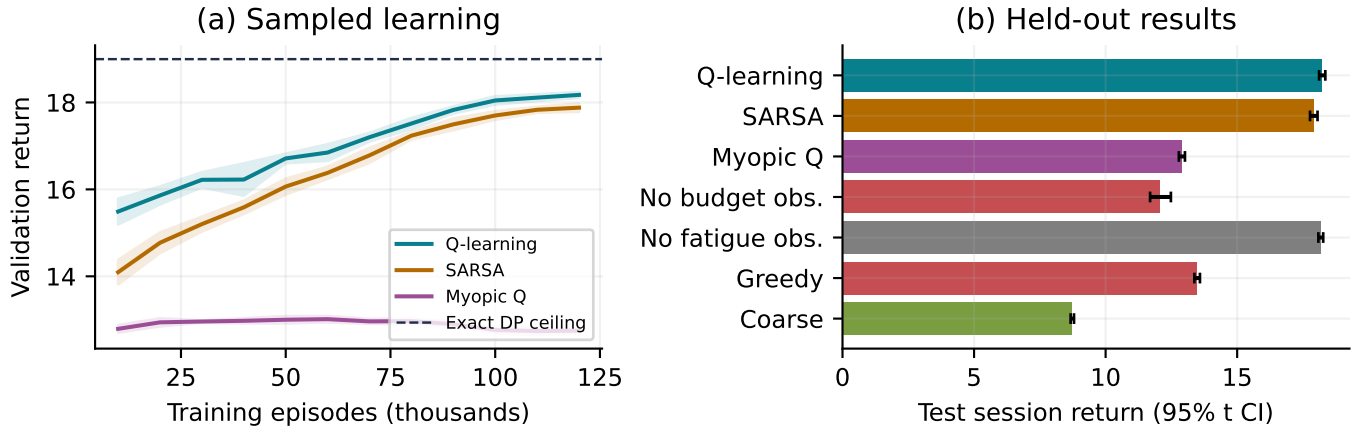


Figure 1: Learning from sampled transitions and held-out comparisons. Shading/error bars are 95% Student-t intervals across eight seed means. The dashed line is the exact expected-return ceiling; validation and test tapes are separate.

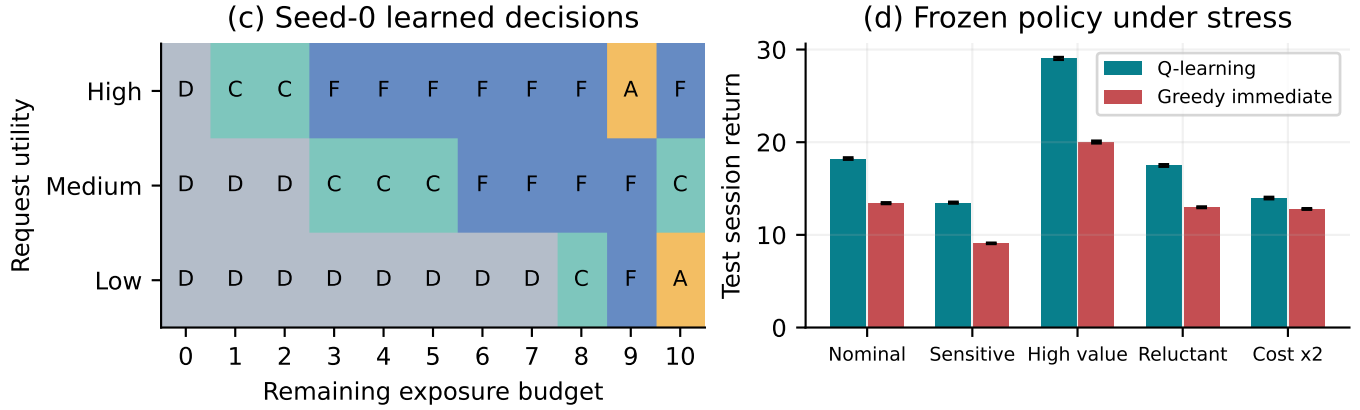


Figure 2: Seed-0 decision slice and frozen-policy stress tests. D/C/A/F denote deny/coarse/ask/full. The slice fixes six requests remaining, medium sensitivity, zero fatigue, and full permission; local boundaries can be noisy. Stress bars show 95% intervals over eight independent seed means.

4 Results, audit, and limitations

Policy	Return ±95% CI	Utility	Exposure
Q-learning	18.23 ± 0.12	23.76	9.48
SARSA	17.91 ± 0.14	23.45	9.61
Myopic Q	12.90 ± 0.12	19.27	9.98
No budget obs.	12.09 ± 0.39	19.26	9.96
No fatigue obs.	18.18 ± 0.09	23.71	9.50
Greedy	13.48 ± 0.11	20.17	9.98
Always coarse	8.73 ± 0.06	14.77	10.00
Exact DP	19.02 ± 0.10	24.52	9.29

Q-learning achieves 18.23 ± 0.12 versus greedy's 13.48 (paired gain 4.75 ± 0.08). It reaches 95.98% of the exact expected optimum 19.00 (estimated gap 0.76). Higher utility with lower exposure than greedy supports better allocation. The sampled DP row differs slightly from its exact expectation due to test noise.

Myopic and budget-blind losses support future planning and budget observation. The fatigue-observation gain is only 0.05 ± 0.06; its interval includes zero, so these runs do *not* establish a reliable marginal fatigue benefit under nominal parameters. SARSA is competitive with a lower measured mean. These results do not establish universal algorithm superiority.

Stress and audit. Frozen Q policies are tested separately on sensitive-heavy and high-value requests, more reluctant consent, and doubled exposure weight. Greedy knows the shifted immediate model (Figure 2). Tests exhaustively verify feasible simulator transitions, budget/permission invariants, reward semantics, deterministic reproduction, and an independent small-MDP Monte Carlo check of exact values. The offline demo uses the saved Q table, displays visit counts, and audits a seeded session. No personal dataset is used.

Validity limits. Exposure units, utility trade-offs, independent requests, and consent probabilities are interpretable assumptions, not user-calibrated measurements. Coarse sharing is abstract; the budget is not differential privacy or a harm probability. Permission is observed. Disclosure composition, adversaries, real data transforms, and long-term user preferences are outside scope. Stress results apply to this synthetic family, not real users. Sparse contexts may have uncertain Q estimates. The small nominal fatigue effect is a useful negative result.

Deliverables. Runnable original code, environment setup, all 40 checkpoints, raw seed metrics, paired intervals, figures, hashes, tests, saved-agent demo, and editable LaTeX/PDF are included. Assistance is documented in the repository provenance note. The supplied GitHub account is labelled at the top; the actual project repository URL must replace it after publication.

References

[1] C. J. C. H. Watkins and P. Dayan. Q-learning. *Machine Learning*, 8:279–292, 1992. Author-hosted paper.

[2] R. S. Sutton and A. G. Barto. *Reinforcement Learning: An Introduction*, 2nd ed. MIT Press, 2018.

[3] H. Nissenbaum. Privacy as contextual integrity. *Washington Law Review*, 79:119, 2004. Publisher record.